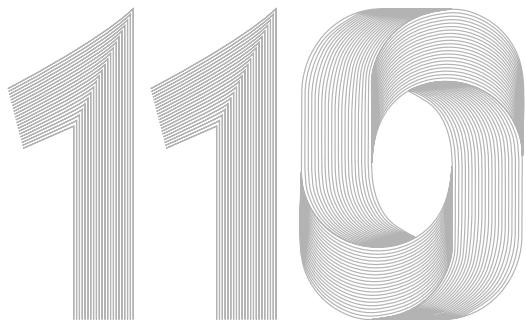




ECOSYSTEMS

Uncovering host diversity and global distribution of *Forcipomyia* Meigen (Diptera: Ceratopogonidae) ectoparasitism on Lepidoptera larvae using citizen science records

CAIO CEZAR D. CORRÊA & CAIQUE D. VASCONCELOS

Abstract: Parasitic interactions play a central role in shaping insect diversity; however, many host-parasite associations remain poorly documented, particularly among Diptera. Species of *Forcipomyia* (*Microhelea*) are known ectoparasites of other insects, yet information on their host associations is still fragmentary. The *Fuliginosa* species-group is known to be ectoparasite of larvae of Lepidoptera and Hymenoptera. In this study, we present the first global synthesis of ectoparasitism by *Forcipomyia* (*Microhelea*), *Fuliginosa* species-group, on caterpillars, integrating literature records with data obtained from the citizen science platform iNaturalist. A total of 109 validated photographic records from 29 countries were compiled, revealing 49 Lepidoptera species belonging to 14 families. When combined with literature records, 15 Lepidoptera families are recognized as hosts, including six reported here for the first time. This interaction shows a broader geographic distribution and higher host diversity than previously recognized, with first records from 17 countries. Our results highlight the potential of citizen science to reveal ecological interactions and demonstrate that parasitism by *Forcipomyia* on Lepidoptera larvae is more frequent, diverse, and widespread than previously documented.

Key words: Biting midges, Citizen Science, iNaturalist, Parasitism, Taxonomy.

INTRODUCTION

The ecological interactions among insects encompass a wide range of relationships (e.g. predation, herbivory, and mutualisms), which together constitute one of the main drivers of biological diversification (Price et al. 2011). Among these interactions, parasitism stands out as one of the most intriguing and complex, as hosts and parasites may interact in multiple ways, involving biocoenotic connections, life cycle synchronization, and host-parasite relationships (Balashov 2011). Despite their high diversity and phytosanitary importance, Diptera (Insecta) still lack detailed information regarding parasitic interactions among their species,

particularly in relation to associations with other arthropods and insect groups (Gillung & Borkent 2017, Marshall & Kirk-Spriggs 2017). Only some taxa show a relatively better resolution of their parasitic habitats, such as Tachinidae (Stireman et al. 2006), Phoridae (Brown et al. 2017, Disney 1986), and Pyrgotidae (Korneyev 2021).

Forcipomyia Meigen, 1818 is the second most diverse genus of Ceratopogonidae, with approximately 1,200 species worldwide, occurring on all continents except Antarctica (Borkent et al. 2025). The subgenus *Forcipomyia* (*Microhelea*) Kieffer, 1917 comprises 106 species (Borkent et al. 2025, Corrêa et al. 2025), many

of which ectoparasite other insect groups (Fig. 1) (Wirth 1956). Wirth (1972) proposed two species-group were based on morphology and host preference. The *Ixodoides* species-group includes species with short and robust mouthparts, specialized in biting Phasmatodea and Orthoptera species (Clastrier & Wirth 1995, Wirth & Castner 1990). The *Fuliginosa* species-group comprises species characterized by slender, elongated mouthparts, that mainly feed on larvae of Lepidoptera and Hymenoptera species (Wirth 1972), with some records on Odonata (Seehausen et al. 2019, Tokunaga 1962). Only three species from the *Fuliginosa* species-group are recorded as ectoparasites of Lepidoptera larvae: *Forcipomyia (Microhelea) eriophora* (Williston, 1896), *Forcipomyia (M.)*

esakiana Tokunaga, 1940, and *Forcipomyia (M.) fuliginosa* (Meigen & Wiedemann, in Meigen, 1818). A total of 19 species of Lepidoptera, in nine families, have been recorded as hosts of species of *F. (Microhelea)*, including Erebididae Leach, 1815 (Edwards 1928), Euteliidae Grote, 1882 (Wirth 1956), Geometridae Leach, 1815 (Baker 1907, Mayer 1937), Notodontidae Stephens, 1829 (Young 1983), Nymphalidae Rafinesque, 1815 (Lever 1939, Salvato et al. 2008, 2012), Papilionidae Latreille, 1802 (Abreu-Rodríguez et al. 2011, Tokunaga 1940, Vasconcelos et al. 2024), Pieridae Duponchel, 1835 (Koptur et al. 2013), Saturniidae Boisduval, 1837 (Mazumdar et al. 2024, Tokunaga 1939), and Sphingidae Latreille, 1802 (Edwards 1923, Knab 1914, Lane 1947, Sevastopulo 1973, Wirth 1956, Wolcott 1948) (Table I).



Figure 1. *Forcipomyia (Microhelea)* biting Lepidoptera larvae. (a) *Battus polydamas* (Linnaeus, 1771), Colombia, ©gregory_nielsen. (b) *Danaus gilippus* (Cramer, 1775), Brazil, ©celiomoura. (c) *Acherontia atropos* (Linnaeus, 1758), South Africa, ©siegwalt. (d) *Parides iphidamas* (Fabricius, 1793), Belize, ©kkrockytop. (e) *Capusa senilis* Walker, 1857, Australia, ©jessat. (f) *Hyphantria cunea* (Drury, 1773), China, ©guanmushan. (g) *Klugeana philoxalis* Geertsema, 1990, South Africa, ©ffwalker. (h) *Ascotis* sp. Hübner, 1825, China, ©liuye.

Table I. List of Lepidoptera larvae species with literature records of ectoparasitism by *Forcipomyia* (*Microhelea*), *Fuliginosa* species-group.

Host	<i>Forcipomyia</i> (<i>Microhelea</i>) species	Country	Reference
Erebidae			
<i>Eudocima phalonia</i>	<i>F. (M.) fuliginosa</i>	Samoa	Edwards (1928)
Euteliidae			
<i>Stictoptera cucullioides</i>	<i>F. (M.) fuliginosa</i>	Indonesia	Wirth (1956)
Geometridae			
<i>Cabera</i> sp.	<i>F. (M.) fuliginosa</i>	Germany	Mayer (1937)
<i>Melanchroia geometroides</i>	<i>F. (M.) eriophora</i>	Cuba	Baker (1907)
Notodontidae			
<i>Lirimiris meridionalis</i>	<i>F. (M.) fuliginosa</i>	Belize	Young (1983)
Nymphalidae			
<i>Anaea troglodyta floridalis</i>	<i>F. (M.) eriophora</i>	U.S.A.	Salvato et al. (2012)
<i>Anaea troglodyta floridalis</i>	<i>F. (M.) fuliginosa</i>	U.S.A.	Salvato et al. (2008)
<i>Doleschallia bisaltide</i>	<i>F. (M.) fuliginosa</i>	Fiji	Lever (1939)
Papilionidae			
<i>Heraclides anchisiades capys</i>	<i>F. (M.) eriophora</i>	Brazil	Vasconcelos et al. (2024)
<i>Papilio demoleus</i>	<i>F. (M.) eriophora</i>	Puerto Rico	Abreu-Rodríguez et al. (2011)
<i>Papilio polytes</i>	<i>F. (M.) fuliginosa</i>	Taiwan	Tokunaga (1940)
Pieridae			
<i>Phoebis sennae</i>	<i>F. (M.) eriophora</i>	U.S.A.	Koptur et al. (2013)
Saturniidae			
<i>Antheraea mylitta</i>	<i>F. (M.) esakiana</i>	India	Mazumdar et al. (2024)
<i>Antheraea mylitta</i>	<i>F. (M.) fuliginosa</i>	India	Mazumdar et al. (2024)
<i>Antheraea pernyi</i>	<i>F. (M.) fuliginosa</i>	China	Tokunaga (1939)
Sphingidae			
<i>Acherontia atropos</i>	<i>F. (M.)</i> sp.	Uganda	Sevastopulo (1973)
<i>Daphnis nerii</i>	<i>F. (M.) fuliginosa</i>	Sri Lanka	Wirth (1956)
<i>Erinnyis ello</i>	<i>F. (M.) fuliginosa</i>	U.S.A.	Knab (1914)
<i>Hippotion celerio</i>	<i>F. (M.) fuliginosa</i>	Samoa	Wirth (1956)
<i>Hyles gallii</i>	<i>F. (M.) fuliginosa</i>	Hungary	Edwards (1923)
<i>Manduca sexta</i>	<i>F. (M.) eriophora</i>	Puerto Rico	Wolcott (1948)
<i>Sphingidae</i> sp.	<i>F. (M.) fuliginosa</i>	Brazil	Lane (1947)

Given the scarcity of specific studies and the ephemeral nature of parasitic interactions, both of which hinder systematic field documentation, citizen science platform emerges as a powerful alternative source of data through photographic and acoustic records (Di Cecco et al. 2021, Manson et al. 2025). The platform iNaturalist has emerged as a rapidly growing source of ecological data with high spatial resolution, enabling the detection of interactions previously invisible to formal science (Prylutskyi & Kapets 2024, Kirichenko et al. 2025).

Recent studies further demonstrate the potential of iNaturalist to investigate host-parasite interactions, insect parasitism, and diverse interspecific associations (Putman et al. 2021, Doherty et al. 2021, De Groot et al. 2024). Based on the observations uploaded to the platform by its users, it is possible to infer biogeographical patterns, estimate the frequency of interaction, and substantially expand the number of available records for neglected taxa. For small insects such as ceratopogonids, macro photography by amateur naturalists has become an unexpected yet extremely valuable source of taxonomic and behavioral information (Gao et al. 2024). Considering the large number of records uploaded to the iNaturalist platform and the few records available in the literature, it becomes evident that there is a substantial opportunity to fill an important gap in understanding of this interaction.

The present study uses a dataset obtained from iNaturalist, together with formal literature, to investigate parasitism by *Forcipomyia* species on Lepidoptera larvae at a global scale. We analyze taxonomic, geographic distribution, and ecological patterns, as well as the diversity of host taxa involved. In addition, we analyze the potential of iNaturalist to reveal ecologically relevant interactions, reinforcing the value of

citizen science for interaction ecology and for undersampled taxonomic groups.

MATERIALS AND METHODS

The collected and validated photographic records were available on the iNaturalist citizen science platform (<https://www.inaturalist.org>). Record searches were conducted using filters and keywords such as 'Ceratopogonidae' and '*Forcipomyia*', as well as geographic terms (countries and continents).

The photographs were examined one by one in order to verify the occurrence of interaction between specimens of *F. (Microhelea)* and caterpillars. Duplicate records of the same specimen photographed and recorded by different users were removed, retaining only one as a validator. Photographs with low-resolution, out-of-focus, poorly positioned, framed, and/or misidentified were discarded. The host identification was made by direct comparison, and refinement was carried out at the lowest possible taxonomic level (order, family, subfamily, tribe, genus, or species). All direct records of *F. (Microhelea)* specimens feeding on caterpillars were validated on the platform, considering observations up to January 01, 2026, and compiled into a table including the following fields: 'Host', 'Country', 'Coordinates', 'iNaturalist Observation', and 'Number of specimens on host' (Table II). Additionally, we compiled data from the literature on all host records involving species or specimens of *F. (Microhelea)* feeding caterpillars (Table I).

The map was created using coordinates from the literature and the platform, with SimpleMappr (<https://www.simplemappr.net>) (Shorthouse 2010). The graph was created using Microsoft Excel®.

Table II. List of Lepidoptera larvae species found on the iNaturalist platform being ectoparasitized by *Forcipomyia* (*Microhelea*), Fuliginosa species-group.

Host	Country	Coordinates	iNaturalist Observation	N
Erebidae				
<i>Agape</i> sp.	Australia	-16.88843, 145.69492	29882650	1
Arctiinae sp.	Bolivia	-17.34357, -63.83720	83391264	2
<i>Asota</i> sp.	Indonesia	-6.33666, 106.70451	27454571	1
Erebidae sp.	Brazil	-24.03048, -47.35056	323352286	1
<i>Hypercompe</i> sp.	Costa Rica	8.64826, -83.46725	198701357	1
<i>Hyphantria cunea</i>	China	40.07647, 116.46345	316275415	1
Orgyiini sp.	Australia	-25.76012, 152.95228	153007387	4
Eupterotidae				
<i>Poloma angulata</i>	South Africa	-28.2294, 32.3935	119602629	2
Geometridae				
<i>Ascotis</i> sp.	China	39.99568, 116.37320	311821693	4
<i>Capusa senilis</i>	Australia	-27.50152, 152.95914	75526905	1
<i>Ectropis excellens</i>	Japan	34.71542, 135.6473	248934340	1
Geometridae sp.	Costa Rica	9.88689, -83.65442	251112975	1
Geometridae sp.	Costa Rica	10.43134, -84.01187	197233872	1
Geometridae sp.	Taiwan	24.19367, 120.79762	1364419461	1
<i>Menophra abruptaria</i>	Portugal	39.49092, -9.15737	294760556	1
Hesperiidae				
<i>Astraptus</i> sp.	Brazil	-22.44262, -44.52693	270417066	1
Hesperiidae sp.	Costa Rica	8.40459, -83.33939	304276013	2
Limacodidae				
<i>Acharia hyperroche</i>	Colombia	4.84385, -75.81872	326213211	3
<i>Acharia</i> sp.	Ecuador	0.15798, -79.06384	324048418	2
<i>Euclea cippus</i>	Brazil	-7.18257, -34.90779	233819855	2
Limacodidae sp.	Brazil	-22.42431, -43.24619	112032191	1
Lycaenidae				
Eumaeini sp.	Ecuador	0.05058, -78.84359	317704297	1
<i>Pseudolycaena marsyas</i>	Bolivia	-17.34338, -63.83841	82185674	2
Noctuidae				
<i>Acyclania tenebrosa</i>	Brazil	-18.32847, -42.75154	128092539	8
<i>Chrysodeixis eriosoma</i>	Guam	13.27968, 144.74853	335042236	1
<i>Klugeana philoxalis</i>	South Africa	-31.39770, 19.14042	325833896	1

Table II. Continued.

Host	Country	Coordinates	iNaturalist Observation	N
Noctuidae sp.	Mexico	14.88936, -92.25245	307600866	2
<i>Spodoptera</i> sp.	Colombia	4.61457, -74.48065	121305880	1
<i>Spodoptera</i> sp.	U.S.A.	25.86681, -80.38121	37085071	4
<i>Xanthopastis timais</i>	Brazil	-18.35308, -42.66135	216643670	1
Nolidae				
<i>Diphthera festiva</i>	U.S.A.	25.39100, -80.52559	322293653	1
Notodontidae				
<i>Neola semiaurata</i>	Australia	-27.50505, 153.08194	273493307	1
<i>Nystalea</i> sp.	Brazil	-7.18257, -34.86125	235099277	1
<i>Phalera flavescens</i>	South Korea	37.58374, 126.80609	107941434	1
<i>Rigema ornata</i>	South Africa	-24.64582, 28.66624	265686159	1
Nymphalidae				
<i>Danaus chrysippus</i>	South Africa	-33.94542, 22.46337	10998664	1
<i>Danaus gilippus</i>	Brazil	-6.35611, -39.27975	226181208	4
<i>Danaus plexippus</i>	Puerto Rico	18.30532, -66.28763	130386770	1
<i>Danaus plexippus</i>	Puerto Rico	18.30540, -66.28779	131338419	1
<i>Danaus plexippus</i>	Cuba	21.08835, -75.93835	36453494	3
<i>Hamadryas</i> sp.	Brazil	-25.33493, -49.11615	208592348	1
<i>Heliconinae</i> sp.	Brazil	-26.93627, -49.09627	21899537	1
<i>Heliconius erato phyllis</i>	Brazil	-18.24093, -42.75753	128699555	1
<i>Heliconius ethilla</i>	Brazil	-19.70506, -44.14814	210123801	4
<i>Heliconius ismenius telchinia</i>	Costa Rica	9.74776, -82.86154	68621018	5
<i>Marpesia petreus</i>	Mexico	20.96814, -89.62987	142828562	2
<i>Mechanitis lysimnia</i>	Brazil	-23.60627, -46.82201	82057947	1
Nymphalidae sp.	Ecuador	-0.22932, -78.37181	215710857	2
<i>Siproeta epaphus trayja</i>	Brazil	-25.44108, -49.23777	331109302	1
<i>Tirumala limniace</i>	Hong Kong	22.43208, 114.37864	137811602	1
Papilionidae				
<i>Battus polydamas</i>	Colombia	4.06030, -73.70037	103264622	5
<i>Heraclides</i> sp.	Mexico	18.58560, -95.07401	154518292	1
<i>Papilio aegaeus</i>	Australia	-27.74649, 153.14812	205794834	1
<i>Papilio agenor</i>	Hong Kong	22.27588, 114.14553	226842718	1
<i>Papilio amynthor</i>	New Caledonia	-20.79759, 165.25134	262101596	2
<i>Papilio demetrius demetrius</i>	Japan	34.76077, 135.69696	139350838	1

Table II. Continued.

Host	Country	Coordinates	iNaturalist Observation	N
<i>Papilio demodocus</i>	South Africa	-25.78102, 28.36338	267355018	1
<i>Papilio machaon</i>	South Korea	36.32482, 126.52780	246903813	1
<i>Papilio protenor</i>	Hong Kong	22.38368, 114.27078	139804260	1
<i>Parides iphidamas</i>	Belize	16.09503, -88.83340	69339527	1
<i>Parides</i> sp.	Panama	9.11847, -79.69805	147985516	1
<i>Phalera xuthus</i>	South Korea	36.79397, 127.21147	185697194	1
Pieridae				
<i>Phoebis philea</i>	Brazil	-26.30319, -48.87588	45918506	1
Saturniidae				
<i>Automeris illustris</i>	Brazil	-23.60627, -46.82201	198416566	9
<i>Citheronia laocoon</i>	Brazil	-16.49571, -49.43161	108987472	1
Hemileucini sp.	Brazil	-20.32978, -40.28678	154198148	1
<i>Holocerina smilax</i>	South Africa	-29.27081, 29.52031	37302151	1
Sphingidae				
<i>Acherontia atropos</i>	South Africa	-25.63361, 28.39823	45707964	4
<i>Acherontia atropos</i>	South Africa	-25.63361, 28.39823	45708122	5
<i>Acherontia atropos</i>	South Africa	-25.63361, 28.39823	45708525	1
<i>Acherontia atropos</i>	Madagascar	-18.06036, 49.36274	326368192	1
<i>Acherontia styx</i>	China	32.06420, 119.01543	315523590	1
<i>Cocytius antaeus</i>	Brazil	-11.34591, -39.66320	226181208	1
<i>Cocytius</i> sp.	Brazil	-18.33986, -42.60952	123628255	1
<i>Daphnis hypothous</i>	India	18.23534, 73.58049	330865986	3
<i>Hyles lineata</i>	U.S.A.	33.91256, -98.51317	45124106	1
<i>Manduca paphus</i>	Brazil	-5.93069, -35.22351	195516975	1
<i>Manduca rustica</i>	U.S.A.	31.77332, -97.58298	322333564	1
<i>Neococytius cluentius</i>	Brazil	-18.39573, -42.63422	128699453	1
<i>Neococytius cluentius</i>	Brazil	-18.31089, -42.75466	128699347	1
<i>Nyceryx</i> sp.	Brazil	-25.33493, -49.11615	256421405	1
<i>Pseudosphinx tetrio</i>	French Guyana	3.61773, -53.21045	70771099	3
Sphingidae sp.	Peru	-12.85310, -69.36764	258823939	2
Sphingidae sp.	Sri Lanka	7.41725, 81.20482	102718396	3
Lepidoptera				
Lepidoptera sp.	Australia	-12.56684, 131.25560	127052122	1

Table II. Continued.

Host	Country	Coordinates	iNaturalist Observation	N
Lepidoptera sp.	Brazil	-8.07287, -35.26693	297416297	1
Lepidoptera sp.	Brazil	-23.63928, -46.61778	306516537	1
Lepidoptera sp.	China	39.80265, 116.35874	325317718	1
Lepidoptera sp.	China	30.359, 120.181	319712622	3
Lepidoptera sp.	China	39.80246, 116.35350	327485079	2
Lepidoptera sp.	China	29.79360, 121.16053	256243238	1
Lepidoptera sp.	China	31.02662, 119.91011	256243236	1
Lepidoptera sp.	China	30.11823, 122.09519	193034934	1
Lepidoptera sp.	China	21.97860, 101.15944	132488036	1
Lepidoptera sp.	Germany	52.40299, 13.73293	240518003	1
Lepidoptera sp.	Hong Kong	22.58293, 114.14890	139690147	1
Lepidoptera sp.	Hong Kong	22.42771, 114.18117	46659299	1
Lepidoptera sp.	India	13.12277, 74.99331	258633739	1
Lepidoptera sp.	India	12.68051, 75.07849	103746555	1
Lepidoptera sp.	Panama	9.31594, -79.63348	102680999	1
Lepidoptera sp.	South Korea	37.56716, 126.88721	107941681	1
Lepidoptera sp.	South Korea	37.56839, 126.88369	95486753	1
Lepidoptera sp.	Singapore	1.44916, 103.72686	150210405	1
Lepidoptera sp.	Taiwan	25.10849, 121.57267	188785248	1
Lepidoptera sp.	Taiwan	24.93850, 121.43339	171721931	1
Lepidoptera sp.	Taiwan	23.84923, 120.84171	111754531	2
Lepidoptera sp.	Taiwan	25.12965, 121.51959	44704113	1
Lepidoptera sp.	U.S.A.	25.56972, -80.34393	269317698	1
Lepidoptera sp.	U.S.A.	25.59119, -80.50823	316031782	1

RESULTS AND DISCUSSION

The search on the iNaturalist platform resulted in 109 records of lepidopteran larvae ectoparasitized by specimens of *F. (Microhelea)* from the Fuliginosa species-group, spanning 29 countries (Fig. 2) across all continents except Antarctica (Table II). Records were predominantly distributed in tropical and subtropical regions, with the highest numbers of observations from Brazil (26), China (10), and

South Africa (9). The countries with the highest numbers of observations were China (506 observations, 0.02%), Brazil (332 observations, 0.07%), and South Africa (144 observations, 0.06%), reflecting both regional biodiversity and uneven observation effort across the platform.

Lepidoptera hosts were identified to the lowest possible taxonomic level based on photographic evidence, resulting in 25 records identified only to order level, 15 to family or

subfamily, 14 to genus, and 55 to species level. In total, representatives of 14 families and 49 species of ectoparasitized Lepidoptera were documented from iNaturalist records (Table II). When these data were synthesized with previously published records, a total of 15

Lepidoptera families is recognized as hosts of *F. (Microhelea)* (Fig. 3). Euteliidae was the only family recorded exclusively in the literature (Wirth 1956), whereas all other families were represented by iNaturalist observations.

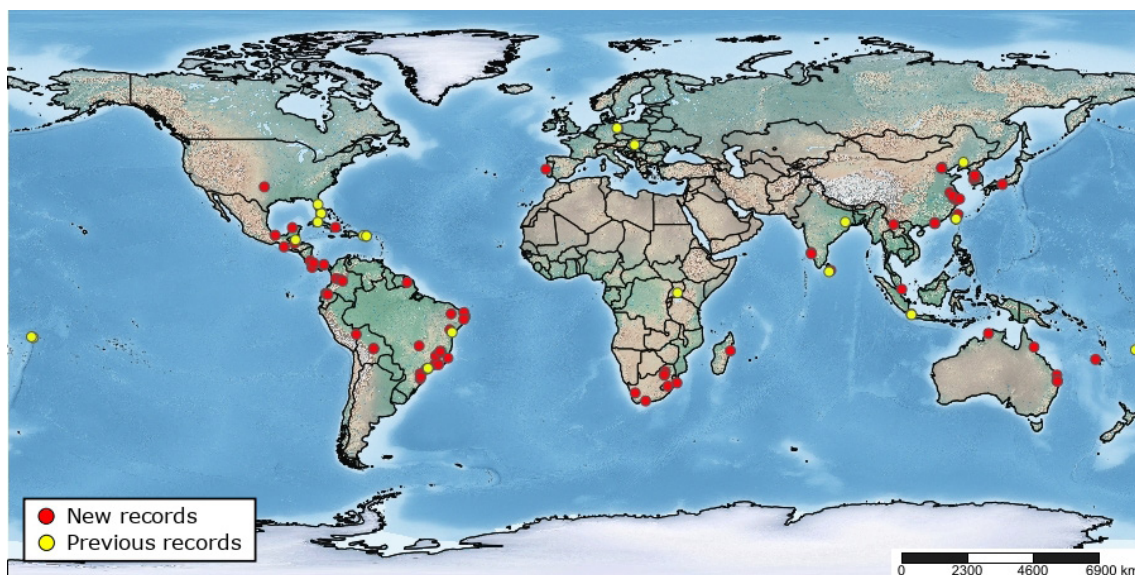


Figure 2. Distribution map of *Forcipomyia (Microhelea)*, Fuliginosa species group, with records of ectoparasitism on Lepidoptera larvae. The yellow circles represent previous records, and red circles represent new records.

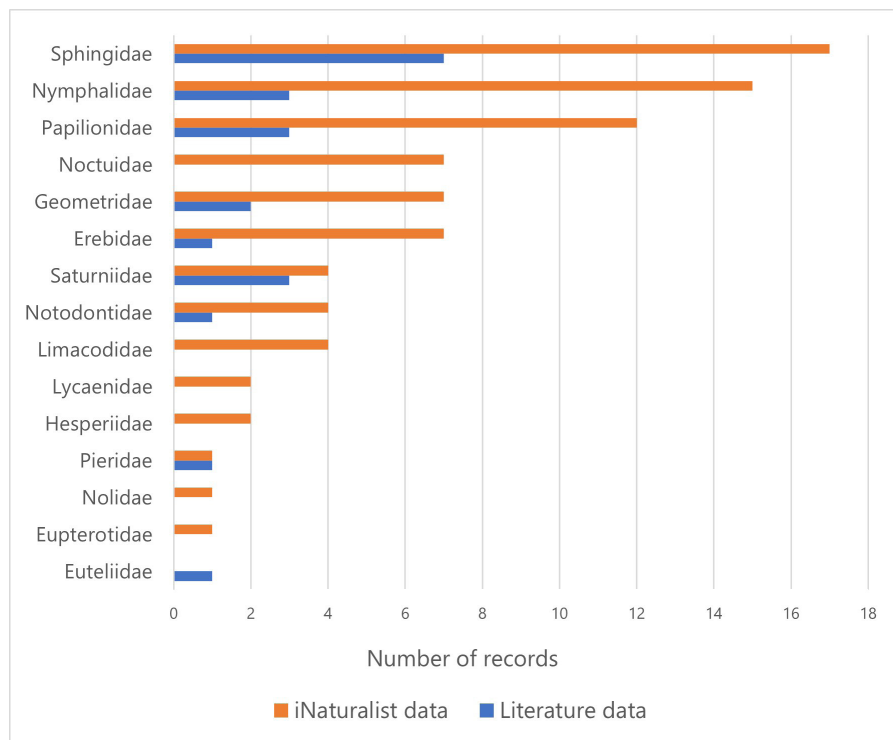


Figure 3. Numbers and sources of records by family of lepidopteran larvae ectoparasitized by *Forcipomyia (Microhelea)*, Fuliginosa species-group.

The families Eupterotidae Swinhoe, 1892, Hesperiiidae Latreille, 1809, Limacodidae Duponchel, 1845, Lycaenidae Leach, 1815, Noctuidae Latreille, 1809, and Nolidae Bruand, 1846 are reported here as hosts of *F. (Microhelea)* for the first time (Table II). The number of ectoparasitism records increased in most families (Fig. 3). Sphingidae, Nymphalidae, and Papilionidae had the highest numbers of records (17, 15, and 12 observations, respectively), suggesting that larger or more visible caterpillars may be detected by observers more frequently.

Parasitism intensity varied considerably among records. Most caterpillars hosted a single *Forcipomyia* individual attached to the integument, however, nine specimens were recorded on *Automeris illustris* Walker, 1855, and eight specimens on *Acyclania tenebrosa* Dognin, 1911. These observations suggest that aggregations may occur on larger, more exposed, or potentially more attractive hosts.

We present the first records of *F. (Microhelea)* as ectoparasites of caterpillars in Australia, Bolivia, Colombia, Costa Rica, Ecuador, French Guiana, Guam, Hong Kong, Japan, Madagascar, Mexico, New Caledonia, Panama, Peru, Portugal, Singapore, and South Africa, bringing the total number of countries with documented presence to 31. Although this interaction had already been documented in North America and the Caribbean (Wirth 1956), the expansion of the dataset reveals a surprisingly wide distribution, spanning tropical, subtropical, and temperate environments on continents represented in iNaturalist, with the exception of Antarctica.

Variation in the taxonomic resolution of host identifications reflects both genuine host diversity and methodological limitations inherent to field photography. In several cases, accurate identification was hindered by viewing

angle, larval instar, or specimen condition. Despite these constraints, the ecological signal remains robust, indicating that *F. (Microhelea)* parasitizes caterpillars across a wide range of sizes, morphologies, and phylogenetic lineages, without apparent restriction to specific Lepidoptera clades.

Citizen science data proved fundamental in revealing these patterns. Platforms such as iNaturalist enable the detection of rare, transient ecological interactions that are often overlooked in traditional entomological surveys (Doherty et al. 2021, Putman et al. 2021). As a user-generated dataset, iNaturalist reflects observation effort rather than true abundance or distribution, consistent with known sampling biases in opportunistic biodiversity data (Kirichenko et al. 2025, Prylutskyi & Kapets 2024). This bias likely contributes to the concentration of records in countries such as Brazil and China, where high biodiversity coincides with large and active user bases.

Taken together, these results demonstrate that integrating literature records, classical taxonomy, and citizen science observations provides a substantially expanded perspective on parasitism by *F. (Microhelea)*. This study represents the most comprehensive synthesis to date of the relationship between *F. (Microhelea)* of the Fuliginosa species-group and Lepidoptera, highlighting the central role of citizen science in uncovering ecologically relevant interactions that have long remained undocumented.

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CCDC: original draft preparation, data collection, identification of Ceratopogonidae specimens, and critical review of the manuscript. CDV: original draft preparation, data collection, identification of caterpillar specimens, and critical review of the manuscript. The authors contributed equally to the development, analysis, and review of the manuscript.

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Data availability

The original data presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

